

Co-work Space Management System

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Interview Outline

Interviewer: Marten atef

Interviewee: Receptionist

Location: Black horse courses, El-Hosary

Appointment Date: 18/3/2026

Start time: 8:15 PM

End time: 9:00 PM

Objectives: Requirements gathering

Topic questions 1: Introduction about company?

Black Horse Courses is a co-working space and training center. It provides courses in fields such as programming, marketing, and design. It also offers a comfortable workspace for students and freelancers.

Topic questions 2: How many departments in a company?

The company has 6 main departments.

Topic questions 3: What is the function of the company's departments?

- **HR Department:** Responsible for hiring and managing employees.
- **Marketing Department:** Promotes courses and attracts new students.
- **Sales Department:** Communicates with customers and sells courses.
- **Instructors:** Deliver training and prepare course content.
- **Customer Service / Reception:** Assists visitors and answers inquiries.
- **IT Support:** Manages technical systems and solves technical issues.

Topic questions 4: Which departments cooperate with each other?

Most departments cooperate with each other. The main cooperation is between Marketing and Sales, Sales and Customer Service.

Topic questions 5: Who is the supplier of the company?

The suppliers include internet service providers, computer equipment companies, and educational content providers.

Topic questions 6: Who is responsible for selling the company's products?

The Sales department is responsible for selling the company's courses and services. The Marketing department supports this process by attracting customers

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Topic questions 1: What is the function of the HR department?

The HR department is responsible for managing employees in general. This includes hiring, organizing contracts, tracking attendance, and also supporting employee development and solving workplace issues.

Topic questions 2: What are the HR departments and their roles in the company?

Inside HR, there are usually different roles such as:

- **Recruitment:** responsible for hiring and selecting candidates
- **Training & Development:** improving employees' skills
- **Payroll:** handling salaries and payments
- **Employee Relations:** managing employee concerns and satisfaction

Topic questions 3: What are the departments that deal with HR within the company?

HR works with almost all departments, but mainly with:

- Management
- Sales and Marketing
- Instructors or trainers for contracts and scheduling

Topic questions 4: How is the employee evaluated?

Employees are evaluated based on:

- Work performance
- Attendance and punctuality
- Achieving targets (especially in Sales)
- Feedback from managers or customers

Topic questions 5: Does the HR department contact external and internal entities?

Yes, HR communicates with external entities such as:

- Training companies (external)
 - Legal or accounting firms (internal)
 - Universities or educational institutions for internships (external)
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Objectives: Requirements gathering from IT department

Topic questions 1: What is the main role of IT department?

The main role of the IT department is to manage and support all technology systems in the company. This includes maintaining computers, networks, software, and ensuring everything runs smoothly.

Topic questions 2: What is your main job as a manager of IT department?

As an IT manager, I am responsible for supervising the IT team, making sure all systems are working efficiently, and solving any technical issues. I also plan upgrades, manage security, and support the company's digital needs.

Topic questions 3: What are the specializations of IT department in the company?

The IT department includes different specializations such as:

- **Technical support**
- **Network administration**
- **System administration**
- **Cybersecurity**

Topic questions 4: Does IT department offer a product or service?

The IT department mainly provides services, not products. These services include technical support, system maintenance, and ensuring smooth digital operations.

Topic questions 5: Who are the IT department suppliers?

IT suppliers include:

- Hardware companies (computers, servers)
- Software providers
- Internet service providers

Topic questions 6: How does IT support the other departments of the company?

IT supports other departments by:

- Fixing technical problems
- Providing systems and software
- Maintaining communication tools

Topic questions 7: Can you show me last month reports and view some of them?

For confidentiality reasons, I'm not allowed to share actual reports.

However, I can show you a sample or explain what these reports usually include, such as system performance, issues handled, and maintenance activities.

Project Documentation: DFD Development Methodology

Steps in Constructing the System Data Flow Diagrams

This section details the methodology used to transform the requirements gathered during the stakeholder interviews into the structured Data Flow Diagrams included in this report. The goal was to ensure that every process and information flow accurately reflects the real-world operations of the co-working space.

We followed a top-down analytical approach, starting with the overall system context and gradually breaking it down into more specific operational details.

Initial Phase: Defining System Scope and External Stakeholders

The first step involved setting a clear boundary for the Co-working Space Management System to distinguish between internal processes and external interactions. By reviewing the interview transcripts, we identified the system as a dual-purpose entity handling both training services and workspace rentals.

We then mapped out the primary external entities that provide data to or receive information from the system. These include the customers, such as students and freelancers who book services, and the instructors who provide the educational content and attendance data. We also identified the suppliers responsible for infrastructure like hardware and internet, the management team that sets the strategic goals and budgets, and various external partners involved in training coordination and legal or financial compliance. This phase resulted in the creation of the Context Diagram.

Second Phase: Analyzing Information Flow and Interaction

Once the boundaries were set, we focused on identifying every specific piece of data moving across the system. This involved a deep dive into the interactions described by the staff to understand exactly how information is exchanged.

For instance, we tracked how a customer's initial inquiry leads to a registration request and eventually a payment, while the system responds with confirmations and schedules. Similarly, we mapped the flow of contracts and payroll data between the system and the instructors. We also accounted for the flow of technical requirements and purchase orders sent to suppliers, as well as the various performance and sales reports delivered to management. This detailed inventory ensured that no critical communication channel was overlooked.

Third Phase: Internal Process Decomposition

After mapping the external interactions, we began breaking down the central system into its core internal functions. Based on the organizational structure described by the reception and management staff, we categorized the system into four primary functional areas.

These areas include Sales and Marketing, which handles lead generation and registrations; Human Resources, which manages the lifecycle of employees and instructors; IT Operations, which oversees technical support and cybersecurity; and Customer Service, which acts as the front-line interface for visitor inquiries. This decomposition allowed us to move from a single "system bubble" to a more detailed view of how different departments actually handle data.

Final Phase: Integration and Systematic Verification

The final step was to integrate all the identified entities, data flows, and internal processes into the Level Zero Diagram. We carefully connected each external stakeholder to the specific internal department they interact with, while also mapping the "behind-the-scenes" data exchange between departments—such as the sales team passing student information to the HR team for instructor scheduling.

To ensure the integrity of the work, we conducted a final cross-reference check against the original interview notes. Every line and process on the diagram was verified to match the documented needs of the business, ensuring that the final DFD is a precise and reliable representation of the Co-working Space Management System.
